

Vaibhav Dhumale

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EDUCATION

Gati Shakti Vishwavidyalaya, Vadodara

BTech in Electronics & Communication Engineering, specialisation in Railway Engineering

CGPA: 7.79

2021- May 2025

Matrix Academy, CBSE School

Class XII, CBSE

Percentage: 93.4%

May 2021

John XXIII CBSE School

Class X, CBSE

Percentage: 90.6%

May 2019

PROFESSIONAL EXPERIENCE

Siemens Technology and Services Private Limited

Graduate Engineer Trainee, Interlocking team Signalling Department

Pune, MH

Aug 2025 - present

- Author for the **Book of Circuit-Interlocking & ETCS** for the MAHSR project and **BIM engineer** for Egypt L1 project.
- Contributed to the design of BOC of 4 Indian Metro Projects India and one of Sydney.
- Gained significant understanding of the design process of interlocking systems with CAD related automations.

Bangalore Metro Rail Cooperation Limited (BMRC Ltd.)

Summer Intern, Signalling and Telecommunication Department

Bangalore, KA

May - June 2024

- Engaged and studied metro operations and signaling, focusing on testing of **CBTC** systems for upcoming projects.
- Contributed to the design review process of **Platform Screen Doors (PSD)**.
- Visited a Metro Depot to learn about maintenance and control procedures.

Research Designs & Standards Organisation (RDSO)

Summer Intern, PS & EMU Directorate

Lucknow, UP

May - June 2023

- Gained insights into RDSO's collaboration with Indian Railways in maintaining standards for railway systems.
- Participated in in-depth learning about the **Train Control And Management Systems**.
- Interacted with Executive Directors of 13 different directorates, understanding their roles in the standards development.

Mumbai Rail Vikas Corporation Limited (MRVC Ltd.)

Summer Intern, General operations

Mumbai, MH

July - August 2022

- Attended training sessions and field visits about railway construction and project management processes.
- Developed an in-depth project titled '**Monitoring Infrastructure Health with the Optical Fibers**'
- Understood MRVC's crucial role in ensuring the efficiency and safety of Mumbai's local railway.

Freelancing roles

Freelancer, Various opportunities.

Remote

2021-present

- CTO | EightWings: Managed and lead the tech team while providing in tech based leads
- Data Analyst | Bintix Ltd.: Effectively used spreadsheets and PowerBi for data management and analysis.
- Video Editor & Graphics Designer | Freelancer.com: Adobe Premiere, DaVinci, Illustrator.
- Assistant Web Developer | Unitybooks: Developed an innovative social media website using the MERN Stack.

PROJECTS

Embedded System based Wagon Fault Detection with ML - [Link]

A system for predictive maintenance

- Developed a robust Raspberry Pi and machine learning based model to detect wagon faults with an interactive GUI.
- Achieved a 97%+ accuracy in training a YOLO model with the results of each train-set stored in reports.
- **Technologies:** Computer Vision, Machine Learning, YOLO, LabelMe, Python3, Tkinter, Django.

MinNotes - [Link]

A feature packed note-taking app

- Developed a robot for RF signal source tracking and obstacle avoidance
- Utilized Arduino, ultrasonic sensors, RF module, and motor shield for adaptive navigation.
- **Skills Applied:** Embedded systems, Fast Fourier Transformation, micro-controller programming, sensor integration.

More Projects on my Portfolio site - [Link]

SKILLS

Languages: English (Fluent), Hindi (Fluent), Marathi (Intermediate)

Programming Skills: Verilog, C++, MATLAB, Python, Dart, MERN Stack, SQL, Flutter, Tailwind-CSS, TensorFlow

Soft Skills: Analytical Thinking, Problem Solving, Project Management, Leadership, Proactive, Adaptive

ADDITIONAL INFORMATION

Certifications: Microsoft Excel VBA and Macros(*Microsoft*), Data Manipulation With PowerBi(*Microsoft*), MATLAB OnRamp (*Mathworks*), Deep Learning (*NVIDIA*), Signalling Engineer Trainee (*IRISET, Hyderabad*)

Interests: Cycling, Photography, Trekking, Reading, Coding, Volleyball, Volunteering

Notable Experiences:

- Founded a pre-seed startup, ReDeFyn, focused on sustainable fashion; pitched at Hult Prize and E-Summit (IITM).